

Mitchell Dickau, PhD | Climate Scientist

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EDUCATION

PhD - Geography, Urban and Environmental Studies

Concordia University, Montréal, QC 2019-2025

- Research on quantifying remaining carbon budgets, modelling Earth system responses to temperature overshoot, and assessing the climate effects of temporary carbon storage.

BA Honours - Geography (Minor in Sociology)

Concordia University, Montréal, QC 2013-2018

- Graduated with distinction, Dean's List
- Honours Thesis: How do weather and climate affect outdoor skating and hockey in Montréal?

WORK EXPERIENCE

Environment and Climate Change Canada

Researcher - Canadian Centre for Climate Services, Data Products Office 2023-2025

- Built an interactive web-app dashboard for visualization of fire weather projections for Canada (<https://climatedata.ca/app/fire-weather-projections/>)
- Processed multi-model climate outputs using Python for clients seeking climate data for adaptation decisions.

Concordia University

Postdoctoral fellow 2026-Present

- Climate modelling studies investigating the climate impact of temporary carbon storage and temperature overshoot.
- Contributor to the *Canada's Changing Climate Report (in prep 2026)*

Undergraduate lecturer 2023

- Undergraduate course (125 students) on the atmosphere and hydrosphere. Topics covered: global energy system, air temperature cycles, weather systems, pollutants, the water cycle, oceans, lakes, and rivers

Teaching Assistant 2019-2025

- Undergraduate courses: Geospatial programming in Python, Scientific Programming in R, Advanced Geographic Information Systems
- Graduate courses: Climate Change and sustainability
- Assisted in teaching labs for geospatial, climate, and environmental data analysis using Python, R, ArcGIS Pro, and TerrSet.

Research Assistant 2018-2019

- Managed the development of an interactive carbon emissions calculator web-app.

Reimagine AI

Consultant 2021-2022

- Supported the development of an AI chatbot meant to discuss climate change

Data Analyst - Terram Vero Consulting

Analyst 2018-2019

- Performed quantitative analysis of mineral sample data and automated data analyses using R.

SKILLS

- Programming: Python (Pandas, NumPy, SciPy, Xarray...), R, Bash, Git
- Geospatial analysis: Python (GeoPandas, Cartopy), ArcGIS Pro, Terrset
- Data Analysis: Interpreting climate data, statistics, climate modelling
- Data visualization: Python (Matplotlib, Panels, HoloViz), R
- Languages: English (fluent), French (advanced)
- Science communication:
 - +10 radio, TV, and print interviews
 - Undergraduate course instructor, and graduate and college level guest lecturer
 - Opinion pieces for The Conversation in French and in English

PEER-REVIEWED PUBLICATIONS

- **Dickau, M.** & Matthews, H. D. (In review, 2026). Temporary land carbon storage can avoid slow-responding climate changes. *Nature Communications*. <https://doi.org/10.21203/rs.3.rs-8725232/v1>
- **Dickau, M.**, Zickfeld, K., & Matthews, H. D. (2026). Irreversible climate changes driven by degree-years of temperature overshoot. *Nature Communications Earth & Environment*. <https://doi.org/10.21203/rs.3.rs-7603499/v1>
- Wynes, S., Davis, S.J., **Dickau, M.** et al. Perceptions of carbon dioxide emission reductions and future warming among climate experts. *Communications Earth Environment*. 5, 498 (2024). <https://doi.org/10.1038/s43247-024-01661-8>
- Wynes, S., **Dickau, M.**, Kotcher, J. E., Thaker, J., Goldberg, M. H., Matthews, H. D., & Donner, S. D. (2022). Frequent pro-climate messaging does not predict pro-climate voting by United States legislators. *Environmental Research: Climate*, 1(2), 025011. <https://doi.org/10.1088/2752-5295/aca8c4>
- **Dickau, M.**, Matthews, H. D., & Tokarska, K. B. (2022). The Role of Remaining Carbon Budgets and Net-Zero CO2 Targets in Climate Mitigation Policy. *Current Climate Change Reports*. <https://doi.org/10.1007/s40641-022-00184-8>
- Matthews, H. D., Zickfeld, K., **Dickau, M.**, MacIsaac, A. J., Mathesius, S., Nzotungicimpaye, C.-M., & Luers, A. (2022). Temporary nature-based carbon removal can lower peak warming in a well-below 2 °C scenario. *Communications Earth & Environment*, 3(1), 65. <https://doi.org/10.1038/s43247-022-00391-z>
- **Dickau, M.**, Matthews, D., Guertin, É., & Seto, D. (2020). Projections of declining outdoor skating availability in Montreal due to global warming. *Environmental Research Communications*, 2(5), 051001. <https://doi.org/10.1088/2515-7620/ab8ca8>

CONFERENCES

- **Dickau, M.** & Matthews, H. D. (May 2025). The proportionality between tonne-years of temporary carbon storage and inertial climate variables. Oral presentation. European Geophysical Union General Assembly, Vienna, Austria
- Matthews, H. D., Zickfeld, K., MacIsaac, A., & **Dickau, M.** (May 2025). Effect of land carbon accounting methods on the climate response to cumulative CO2 emissions. Oral presentation. European Geophysical Union General Assembly, Vienna, Austria.

- Matthews, H. D., Luers, A. & **Dickau, M.** (December 2024). Application of tonne-year accounting to track the durability and climate impact of temporary land carbon. American Geophysical Union, Washington DC, USA.
- **Dickau, M.** & Matthews, H. D. (May 2024). Investigating the path dependence of climate changes in a 200-member ensemble of overshoot scenarios. (Poster) European Geophysical Union General Assembly, Vienna, Austria.
- Matthews, H. D., Zickfeld, K., **Dickau, M.**, MacIsaac, A., Mathesius, S., Nzotungicimpaye, C-M. & Luers, A. (May 2022). Temporary nature-based carbon removal can lower peak warming in a well-below 2°C scenario. (Oral presentation) European Geophysical Union General Assembly, Vienna, Austria.
- **Dickau, M.** (March 2020). Are the planned rates of CO₂ sequestration in the IPCCs aggressive mitigation scenarios achievable and sustainable? (Poster). Sustainability and the Climate Crisis, Montreal, Canada.
- **Dickau, M.** & Matthews, H. D. (June 2019). Projections of declining outdoor skating availability in Montreal due to global warming. Oral presentation. 27th International Union of Geodesy and Geophysics General Assembly, Montreal, Canada.

SELECTED ACHIEVEMENTS

- Awarded Graduate Research Communicator of the Year, Concordia University (2024)
- Recipient of National Science and Engineering Research Council (NSERC) Doctoral Scholarship (2021-2024)
- Fonds de recherche du Québec - Nature et technologies (FRQNT) Master's Scholarship (2019)
- Concordia Faculty of Arts and Science Graduate Fellowship Scholarship - Masters (2019) and Doctoral (2020-2024)
- Hydro-Quebec Master's Scholarship
- Concordia Top 1% Scholar - Faculty of Arts and Science (2017)